Science Olympiad
Materials Science C Test
UT Austin Fall 2025 Invitational

Team Name: ______________________________

Team Number: ______________________________

Directions:

- You will have 50 minutes to complete the exam
- Point values are indicated for all questions along with tiebreaker order
- The maximum number of obtainable points on this exam is 101.
- You may take apart the test to split the work, but make sure to staple it back together at the end of the event.

For grading use only

Page:	2	3	4	5
Points:	13	13	12	19
Score:				
Page:	6	7	8	Total
Points:	16	23	2	98
Score:				

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Problem 1 (15 points)

Dr. Pym shrinks you down to the µm scale using his Ant Man suit. He places you on top of two pieces of cold-worked copper: One heated to 350°C, allowing recrystallization to occur while the other was heated to 500°C for a significantly longer time, causing grain growth to occur What are some differences that you may notice between the two cold-worked copper samples?

2. (3 points) Below is the Hall-Petch equation. Describe what each symbol/letter represents and their units using fundamental metric SI units.
   $\sigma_y = \sigma_o + \frac{k_y}{\sqrt{d}}$

3. (1 point) Based on your observations on both samples, which sample do you expect to have a higher σ_y value? A. Recrystallized copper B. Grain-growth copper

4. (1 point) Why does this higher σ_y value occur?

5. (4 points) Graph σ_y as a function of d for the Hall–Petch Equation given. Does the Hall–Petch relation apply for all values of d? If not, which values does it apply to?

6. (2 points) Which effect does the line drawn exhibit? Label the line accordingly.

7. (2 points) Draw the Reverse Hall–Petch effect line on the graph and label it accordingly.

[Student should draw the line on back of exam sheet 4.]

Problem 2: Professor Oak's Pokéballs (25 points)

Professor Oak wants to first make a Pokéball that won't break upon hitting rock- or steel-type Pokémon, but can also resist heat from fire-type Pokémon.

8. (4 points) Explain two reasons why a ceramic nanomaterial is best for this purpose.

9. (1 point) **Bonus:** Give this Pokéball a new name.

Kurt is considering building a Pokeball factory in his hometown of Azalea Town. He wants to use Bottom-Up Production. Explain why this is advantageous for each of the following:

10. (2 points) Keeping Azalea Town clean (think waste).

11. (2 points) Energy efficiency; Azalea Town has little access to good energy sources?

12. (2 points) Precisely building Pokéballs?

The Devon Corporation has asked Kurt to develop a brand-new, cutting edge Pokeball called the Voltball. It will have improved efficiency for catching electric-types, and glow when a small current is applied from an electric-type pokemon caught inside. To do this, Kurt plans to use Electroluminescent Coatings.

13. (2 points) If he wants the Pokeballs to glow red, would he want a smaller or larger bandgap?
A. Smaller B. Larger C. Depends on the size of the Pokéball D. Cannot determine

14. (1 point) If he wants another class of Voltballs to instead glow yellow, would he have to increase or decrease the nanoparticle size?
A. Increase B. Decrease C. Nanoparticle size does not affect color D. Cannot determine

15. (1 point) What change in wavelength does this correspond to?
A. An increase B. A decrease C. It stays the same D. Cannot determine

Team Rocket is developing a Pokeball that can be placed in poisonous gas in order to poison the Pokemon inside, but still resist degradation from chemical reactions. Their lead scientist is considering Metal-Oxide Nanoparticles.

16. (2 points) How does the high surface area to volume ratio of these nanoparticles allow the gas to linger inside the Pokeball for longer?
    A. Semi-permeable so gas goes in but not out
    B. Larger → attracts more gas but harder to leave
    C. More absorption sites slow diffusion
    D. None of the above

17. (4 points) (Tiebreaker #1) How can the scientist use the composite effect with carbon-based nanomaterials to control diffusion?

The Weather Institute in Hoenn is partnering with the Devon Corporation to build weather-resistant Pokeballs to deal with the desert in Hoenn.

18. (2 points) Why would small nanoparticles be a good option to reflect UV light and prevent overheating?

19. (2 points) Why would materials with good thermal conductivity prevent uneven expansion?

Problem 3: Shuri's Vibranium Alloys (35 points)

Shuri discovers a new Vibranium alloy, called 'Ultra-Vibranium.' Ultra-Vibranium is even stronger than regular Vibranium, and is similar in structure to diamond.

20. (3 points) Describe why diamond, which has sp^3 hybridization, has poor conductivity but is extremely durable

When testing the Ultra-Vibranium , T'Challa exerts a force of 500 N pulling apart a 1 meter cylindrical rod of New Vibranium with radius 3 cm, which makes it stretch by 177 nm. What is the elastic modulus of Ultra-Vibranium?

21. (3 points) What is the elastic modulus of Ultra-Vibranium?
    A. 1×10^{12} Pa
    B. 1×10^{9} Pa
    C. 1×10^{6} Pa
    D. 1×10^{3} Pa

Shuri is considering the thermal conductivity of Ultra-Vibranium.

22. (2 points) Does diamond tend to have high or low thermal conductivity? A. High B. Low

23. (3 points) What is a disadvantage of low thermal conductivity in the case of an explosion?

Assume building Ultra-Vibranium is the same process as building diamond. Shuri considers High Pressure High Temperature (HPHT) and Chemical Vapor Deposition (CVD) methods of forming the material. Which one would:

24. (1 point) Allow her to make layered Ultra-Vibranium? A. HPHT B. CVD

25. (1 point) Produce purer Ultra-Vibranium? A. HPHT B. CVD

26. (1 point) Create larger Ultra-Vibranium? A. HPHT B. CVD

Shuri now finds another Vibranium alloy, called Lightning Vibranium, that is similar to graphene. She plans to use Lightning Vibranium for weapons that can electrocute because of its excellent conductivity.

27. (5 points) (Tiebreaker #2) Explain why Lightning Vibranium, which has an sp^2 hybridized honeycomb lattice, is a much better conductor than Ultra-Vibranium.

Shuri is working with a scientist to make Ultra-Vibranium and Lightning Vibranium less visible in low light but still reflect Infrared Radiation so that they don't heat up excessively. (8)

28. (2 points) Why would a shiny metal mirror coating not work?
    - A. Absorbs too much light
    - B. Reflects too much light
    - C. Always too hot
    - D. None of the above

29. (3 points) Identify one way they can make the Vibraniums reflect IR Radiation but absorb visible light.

30. (3 points) For your answer in ii., critique how this would affect the material durability.

Shuri creates a third new vibranium, called Bane Vibranium, which can be used with a new poison that she has discovered, called Mortivex. Bane Vibranium can be dipped in Mortivex to make it even more deadly.

31. (2 points) For the Bane Vibranium to efficiently absorb the poison, would she want a high (>90°) or low (<90°) wetting angle between Mortivex and Bane Vibranium.? A. High (>90°) B. Low (<90°)

32. (2 points) For the Bane Vibranium to repel water, would she want a high (>90°) or low (<90°) wetting angle between water and Bane Vibranium. A. High (>90°) B. Low (<90°)

33. (4 points) (Tiebreaker #3) Assume Mortivex has low surface tension compared to water, and is nonpolar. Explain how a surfactant with a hydrophobic tail and a hydrophilic head would allow for the appropriate wetting angles from the past two problems.

Problem 4: Lab – Hardness of Aluminum (25 points)

You are given 3 aluminum samples, each having undergone a different heat treatment: 2 hour aging, quenching, and annealing. In this lab you will identify which sample has undergone which treatment and explain why each treatment had the effects that it had

Lab Procedure:

1. Place strip 1 on the table.
2. Place the spring punch on the sample and press to make an indent. Perform 5 indents.
3. Record indent diameters using a caliper.
4. Repeat for strips 2 and 3.
5. Calculate average indent diameter (D_{avg}) for each strip.
6. Compute hardness proxy: $H = \frac{1}{D}$.

34. (3 points) Calculate the hardness proxy using the given formula for indent on each sample.

35. (5 points) Plot hardness proxy vs indent diameter samples and draw the subsequent trendline.

[Student should create plot on back of exam sheet 4.]

36. (6 points) Which sample has undergone which treatment?

37. (4 points) Assuming the Hall–Petch Effect applies, which sample has the largest grain size? Which has the smallest? Why?

38. (5 points) Where would an overaged sample go on the hardness proxy chart? Draw and label the point and explain why it would be there.

39. (2 points) Explain why the hardness proxy is a useful tool in this lab, even if it does not give absolute hardness values in standard units.

Space for Problem 5-7 Chart:

Space for Problem 35 Chart: